

An Exact Weighted-Cycle Obstruction for the 812-Vertex AGL(1, 29) Orientation Instance

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Abstract

Garcia isolated an orientation problem on a deposited 812-vertex, girth-14 cubic Cayley graph of AGL(1, 29) and reported the instance as undecided. At every vertex one must select an incident edge, subject to an upper bound of $3\ell - 33$ selections lying on each simple cycle of length $\ell \in \{14, 15, 16\}$. We prove that no such selection exists. The proof is an exact certificate assigning positive integer weights to 1,172 cycles of length 14 and 12 cycles of length 16. The constraints require a weighted total of at least 16,216,704 off-cycle selections, whereas independent maximization at the 812 vertices permits at most 15,153,552. The strict gap is verified by integer arithmetic directly against the frozen graph6 record. This conclusion concerns only the specified base graph; it does not rule out other bases or resolve the general H_{15} construction.

1 The fixed instance and its scope

Let B be the labelled graph encoded by the unique graph6 record `agl_29_g14.g6` in the data archive accompanying Garcia [2]. The record defines a connected cubic graph with 812 vertices, 1,218 edges, and girth 14. Garcia’s window calculus for the H_{15} replacement reduces avoidance of a 64-cycle on this base to the following incident-edge selection problem; see in particular Definition 3.1, Lemma 3.2, and the discussion following Lemma 5.2 of [2].

For every $v \in V(B)$, select one edge $s(v)$ incident with v . If C is a simple cycle, put

$$a_s(C) = |\{v \in V(C) : s(v) \in E(C)\}|.$$

The question is whether a selection s can satisfy

$$a_s(C) \leq 3\ell - 33 \tag{1}$$

for every simple ℓ -cycle of B and every $\ell \in \{14, 15, 16\}$. Thus the three capacities are 9, 12, and 15.

The closest primary source reported that the corresponding orientation instances for $p = 29, 31, 37$ remained undecided after bounded computation [2, Section 5]. Our contribution is an exact negative certificate for the deposited $p = 29$ record. A bounded primary-source search on 9 September 2026 found no later version of that paper or separate exact resolution under the searched identifiers and phrases; this is a description of the search, not a claim of priority or exhaustive literature coverage.

The use of a compact, exactly replayable artifact follows a broader proof-carrying-computation principle. As a limited methodological comparison, Zeng [3] also separates non-finite reasoning from finite comparisons certified by exact arithmetic in a cycle-length problem. That paper supplies no premise for the theorem below and is not prior work on this graph instance.

Theorem 1. *For the labelled graph B encoded by `agl_29_g14.g6`, there is no incident-edge selection s satisfying (1) for every simple cycle of every length $\ell \in \{14, 15, 16\}$.*

2 A weighted-cycle obstruction

We first give the elementary inequality behind the certificate. It applies to any cubic graph and does not depend on a SAT solver.

For a simple cycle C in a cubic graph and a vertex $v \in V(C)$, let $\bar{e}_C(v)$ denote the unique edge incident with v that is not an edge of C . Given a selection s , define

$$o_s(C) = |\{v \in V(C) : s(v) = \bar{e}_C(v)\}|.$$

The two cycle edges and the one remaining edge partition the three choices at each cycle vertex, so

$$o_s(C) = |C| - a_s(C). \quad (2)$$

In particular, (1) implies

$$o_s(C) \geq 33 - 2|C|. \quad (3)$$

Lemma 2 (Weighted-cycle inequality). *Let G be cubic, let $\mathcal{F}$ be a finite family of simple cycles, and assign a nonnegative integer weight w_C to every $C \in \mathcal{F}$. For a vertex v and an edge e incident with v , set*

$$\lambda(v, e) = \sum_{\substack{C \in \mathcal{F}: v \in V(C) \\ \bar{e}_C(v) = e}} w_C.$$

Every incident-edge selection satisfying (3) on $\mathcal{F}$ obeys

$$\sum_{C \in \mathcal{F}} w_C (33 - 2|C|) \leq \sum_{v \in V(G)} \max_{e \ni v} \lambda(v, e). \quad (4)$$

Consequently, a strict reverse inequality certifies infeasibility.

Proof. Multiply (3) by w_C and sum over C . Reversing the finite order of summation gives

$$\sum_{C \in \mathcal{F}} w_C (33 - 2|C|) \leq \sum_{C \in \mathcal{F}} w_C o_s(C) = \sum_{v \in V(G)} \lambda(v, s(v)).$$

Each vertex makes one of its three choices independently, hence $\lambda(v, s(v)) \leq \max_{e \ni v} \lambda(v, e)$. Summing these inequalities proves (4). $\square$

The last maximum is deliberately taken separately at every vertex. Thus it relaxes nothing about the original decision variables: the original problem also permits each vertex to choose any one of its three incident edges, with no consistency condition between the choices at the two ends of an edge.

3 The integer certificate

The accompanying file `exact_cycle_weights.tsv` lists one positive integer weight and one canonical labelled cycle on each data row. Its support and weight sums are shown in table 1.

Table 1: Support of the weighted-cycle certificate.

cycle length	supported cycles	sum of weights
14	1,172	3,243,141
15	0	0
16	12	999

For a 14-cycle, (3) requires at least 5 off-cycle choices; for a 16-cycle it requires at least one. The left side of (4) is therefore

$$5(3,243,141) + 1(999) = 16,216,704. \quad (5)$$

Exact accumulation of the 2,436 values $\lambda(v, e)$, three for each of the 812 vertices, gives

$$\sum_{v \in V(B)} \max_{e \ni v} \lambda(v, e) = 15,153,552. \quad (6)$$

The per-vertex maxima range from 18,488 to 19,015. Even the coarser uniform estimate

$$\sum_v \max_{e \ni v} \lambda(v, e) \leq 812(19,015) = 15,440,180$$

is strictly smaller than (5).

Proof of theorem 1. Assume that a selection s satisfies all the stated constraints. It then satisfies them on the 1,184 positively weighted cycles in the certificate. By Lemma 2, (5) cannot exceed (6). In fact it exceeds it by 1,063,152, a contradiction. $\square$

Only a subset of the constraints is needed for the contradiction. In particular, neither completeness of a census of all short cycles nor the absence of 15-cycles is a logical premise of the proof: any hypothetical solution of the full instance would necessarily satisfy every constraint in the displayed support.

4 Exact verification and reproducibility

The certificate is finite but too long to print usefully. The principal Python checker supplied with it performs the following steps using Python integers throughout.

1. It extracts and decodes `ag1_29_g14.g6` from the deposited archive, and verifies 812 vertices, 1,218 edges, degree three, and girth 14.
2. For every certificate row, it checks a positive integer weight, a simple cycle of the claimed length in the fixed labelled graph, and the unique non-cycle edge at every cycle vertex.
3. It accumulates the three values $\lambda(v, e)$ at every vertex and recomputes the two sides of (4), including the strict gap.

From the project root, the principal replay is

```
python3 evidence/check_exact_weight_certificate.py \
  ../../batches/literature/bigMac-19/supplement.zip \
  ag1_29_g14.g6 evidence/exact_cycle_weights.tsv source.md \
  evidence/exact_weight_check.json
```

The data archive is also available at the DOI in [1].

A second implementation, `replay_exact_weight_certificate.py`, replays the same weights from the independently serialized cycle census and direct CSP incidence table. It obtains the same lower bound, upper bound, and gap. A fresh verifier written for mathematical review reconstructed the graph from the serialized cycle edges, recovered the graph6 record, and independently obtained 2,030 cycles of length 14, no cycles of length 15, and 5,278 cycles of length 16. These redundant census facts test the data binding; they are not substitutes for the inequality proof.

The integer weights were found by rounding the nonnegative cycle component of a floating-point HiGHS dual ray. The ray, the rounding heuristic, earlier SAT timeouts, and solver status play no role in theorem 1. Once the integer table is given, verification consists only of graph checks, cycle checks, addition, multiplication, comparison, and the proof of Lemma 2.

5 Limitations

The theorem rejects the single deposited AGL(1, 29) base and its complete length-14–16 orientation CSP. It says nothing about the corresponding $p = 31$ or $p = 37$ bases, about non-Cayley bases, or about every possible H_{15} replacement. In particular it neither disproves the broader H_{15} route nor resolves the Erdős–Gyárfás conjecture. The bounded literature screen establishes only that the closest inspected source left this instance undecided; it does not establish priority.

Computational and AI assistance disclosure

An OpenAI Codex system assisted with proof search, exact-checking code, and manuscript preparation. A floating-point optimization run was used only to discover candidate integer weights. The published implication depends solely on the explicit integer certificate and deterministic exact replays described above; a separately written exact verifier was also used during mathematical review. These checks do not replace external peer review, and the author is responsible for the statement, proof, and scope.

Data availability

The fixed graph is in Garcia’s public data archive [1]. The accompanying publication materials contain the integer weight table, two replay programs, the canonical short-cycle census, the direct incidence table, and their machine-readable reports. No proprietary solver or floating-point file is required to verify the theorem.

References

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- [3] Z. Zeng. “The Distinct-Cycle-Length Probability is Strictly Decreasing after 30” (2026). DOI: 10.2139/ssrn.7380519. URL: <https://doi.org/10.2139/ssrn.7380519>.